

From physical GBZ formulations to polynomial roots

A numerical bridge for amoebic and strip GBZs

pyGBZ2d research project

2026-09-07

1.1 Complex momentum describes skin localization

$$\psi_{n_1, n_2} \propto \beta_1^{n_1} \beta_2^{n_2}, \quad \beta_j = e^{\mu_j + i\theta_j}$$

Phase and envelope

θ_j controls spatial oscillation. μ_j controls exponential growth or decay along a lattice direction.

A GBZ selects the complex momenta compatible with the boundary problem.

The 2D question

Which pairs of localization exponents and phases survive when boundaries are opened?

Amoebic and strip constructions organize this selection in different ways.

1.2 Amoebic GBZ: select localization in modulus space

Write $f(E, \beta_1, \beta_2) = \det[EI - h(\beta_1, \beta_2)]$.

$$\mathcal{A}_E = \{(\ln|\beta_1|, \ln|\beta_2|) : f(E, \beta_1, \beta_2) = 0\}$$

Project away the phases

The amoeba records where a reference energy can be supported by complex Bloch waves with given envelopes.

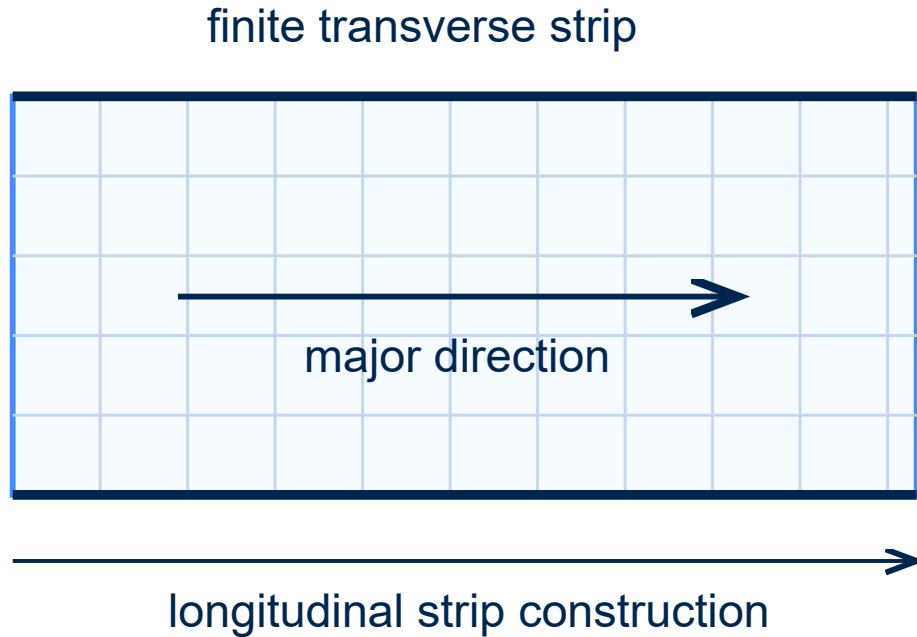
The Ronkin minimum selects the logarithmic radii; the GBZ retains the corresponding zeros and their phases.

Central-hole criterion

A central zero-free hole gives a flat Ronkin minimum. Its disappearance identifies the amoeba spectral support.

Finding a minimum and finding a GBZ point are distinct questions.

1.3 SGBZ: resolve the boundary problem by a strip



Geometry schematic. No lattice parameters or data implied.

Keep a preferred direction

Open the transverse direction and treat the strip as a quasi-1D system along its major axis.

Its thermodynamic winding selects the longitudinal radius. Transverse modes obey the 1D GBZ condition.

The chosen major direction carries physical information about the boundary geometry.

1.4 Two physical prescriptions, one numerical objective

Amoebic GBZ

Select radii through a convex spectral potential in the two-dimensional modulus plane.

SGBZ

Select radii through a direction-dependent strip boundary problem.

Project objective

Translate each prescription into roots of a one-variable polynomial, while retaining its own physical selection rule.

Compare spectra and localization on a common representation.

The SGBZ theory gives spectral inclusion within the amoeba spectrum; it does not equate the two prescriptions for every geometry.

2.1 Fix one Bloch factor, solve for the other

$$f(E, \beta_1, \beta_2) = \sum_{n=-M}^N c_n(E, \beta_1) \beta_2^n$$

$$P(\beta_2) = \beta_2^M f = c_N \prod_{j=1}^{M+N} (\beta_2 - \beta_{2,j})$$

Scan one angle

Fix E and μ_1 , then let $\beta_1 = e^{\mu_1 + i\theta_1}$ traverse a circle.

Solve $P = 0$ to obtain the transverse modes at each angle.

Retain the root geometry

$$\ell_j(\theta_1) = \ln |\beta_{2,j}(\theta_1)|$$

Moduli encode localization; arguments retain the phases needed to reconstruct the GBZ.

Generic finite nonzero roots and $c_N \neq 0$ shown here; degree loss and multiple roots require separate treatment.

2.2 Amoeba bridge I: integrate one phase analytically

$$R_E(\mu_1, \mu_2) = \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} \ln|f| \, d\theta_2 \, d\theta_1$$

Jensen's identity for each root

$$\frac{1}{2\pi} \int_0^{2\pi} \ln|e^{\mu_2 + i\theta_2} - \beta_{2,j}| \, d\theta_2 = \max(\mu_2, \ell_j)$$

$$R_E = \frac{1}{2\pi} \int_0^{2\pi} \left[\ln|c_N| - M\mu_2 + \sum_j \max(\mu_2, \ell_j(\theta_1)) \right] d\theta_1$$

The two-phase logarithmic integral becomes a one-angle expression in polynomial roots.

Root-factorization/Jensen reduction of the Ronkin definition; all coefficients and roots depend on E and β_1 .

2.3 Amoeba bridge II: stationarity becomes root counting

$$a_2 = \partial_{\mu_2} R_E = \frac{1}{2\pi} \int_0^{2\pi} [\#\{j : \ell_j(\theta_1) < \mu_2\} - M] d\theta_1$$

Only crossings change the count

Solve $\ell_j(\theta_1) = \mu_2$. Between crossing angles, the number of enclosed roots stays constant.

$$a_2 = \sum_a (n_a - M) \frac{\Delta\theta_{1,a}}{2\pi}$$

Select the Ronkin minimum

Search for $a_2 = 0$, then impose $a_1 = 0$ through the outer radius search.

Exchanging axes gives the corresponding root-count identity for a_1 .

Derivative formula applies at differentiable points; flat tracks and singular minima need one-sided or subgradient conditions.

2.4 SGBZ bridge I: transverse OBC becomes a root tie

$$\left|\beta_{2,(1)}\right| \leq \dots \leq \left|\beta_{2,(M+N)}\right|$$

The transverse 1D GBZ condition

$$\left|\beta_{2,(M)}\right| = \left|\beta_{2,(M+1)}\right|$$

Physical balance

The two middle modes have equal exponential weight, allowing the transverse OBC bulk-band condition.

Numerical condition

Find zeros of the middle-root gap:

$$g(\theta_1) = \ell_{(M+1)}(\theta_1) - \ell_{(M)}(\theta_1)$$

C. Wang et al., arXiv:2506.22743v2, Sec. II; parenthesized indices denote modulus rank, with multiplicity.

2.5 SGBZ bridge II: replace the strip by loop winding

$$\mu_{2,\text{mid}}(\theta_1) = \frac{\ell_{(M)}(\theta_1) + \ell_{(M+1)}(\theta_1)}{2}$$

Between middle roots, this reference encloses exactly M roots and cancels the transverse Laurent pole.

$$w(\theta_2) = \frac{1}{2\pi} \int_0^{2\pi} \Im \left[\frac{1}{f} \frac{df}{d\theta_1} \right] d\theta_1$$

$$W(E, \mu_1) = \frac{1}{2\pi} \int_0^{2\pi} w(\theta_2) d\theta_2 = \sum_a w_a \frac{\Delta\theta_{2,a}}{2\pi}$$

Middle-pair events partition the phase circle into intervals of constant loop winding.

2.6 SGBZ selection becomes a scalar radius problem

Longitudinal selection

$$W(E, \mu_1^*) = 0$$

Find the localization radius

Vary μ_1 and evaluate the root-based winding.
A bracketed scalar search locates its zero.

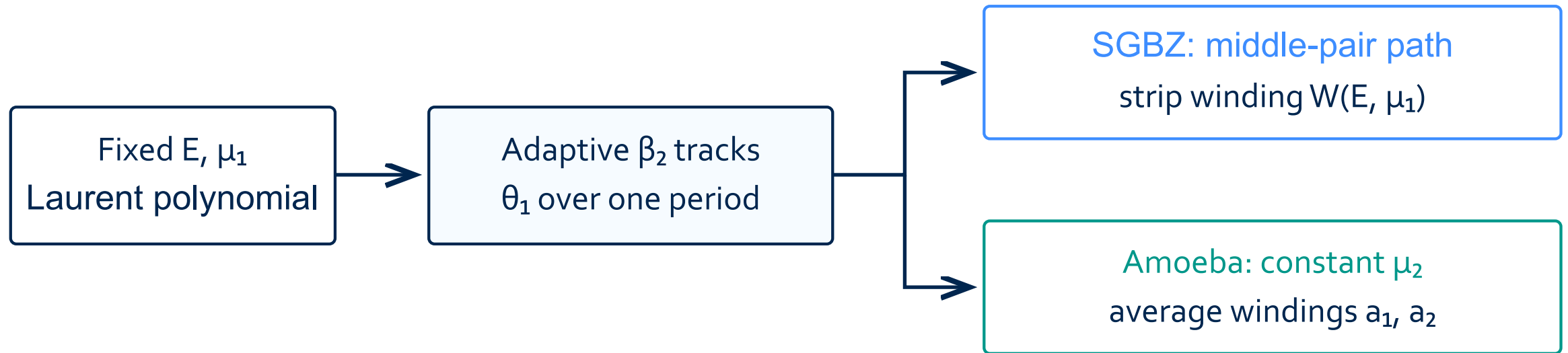
Persistent middle-root ties require one-sided winding limits.

Recover the Bloch waves

At the selected radius, retain the middle-pair roots and their phases.

A zero-winding plateau still requires a spectral-membership check.

3.1 Shared roots, different physical constraints



Amoeba: constant level

Intersections with a trial μ_2 determine Ronkin-gradient constraints.

SGBZ: moving middle gap

Middle-root ties and loop winding determine the strip constraint.

One representation serves both constructions without identifying their physical selection rules.

3.2 Why follow roots rather than sample a dense grid?

Preserve connected modes

Root continuation retains branch identity and phase as the longitudinal angle changes.

Root identity and instantaneous modulus rank are different notions.

Resolve changes where they occur

Adapt the sampling near crossings and degeneracies. Refine the events that change counts or winding.

Between events, use the constant topological information.

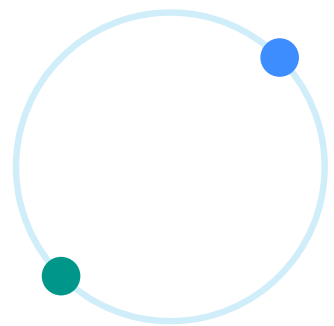
Design advantage

Reduce singular quadrature and broad parameter scans to polynomial solves, event locations, and scalar constraints.

This is a structural numerical advantage; quantitative speed and accuracy gains require measured convergence tests.

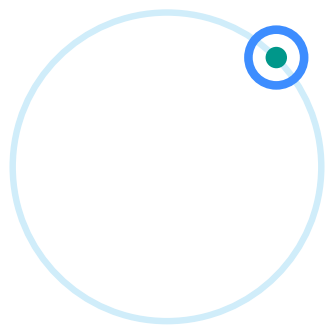
3.3 Isolated modes, degeneracies, and continuous families

Isolated modulus tie



$|\beta_a| = |\beta_\beta|$
 $\beta_a \neq \beta_\beta$

Multiple root



$\beta_a = \beta_\beta$
 $\partial f / \partial \beta_2 = 0$

Continuum in modulus



equality over an interval

SGBZ: tied tracks
Amoeba: flat log modulus

Conceptual event taxonomy. Symbols and line pairs are schematic, not computed root trajectories.

Equal decay lengths need not mean coincident complex Bloch factors. Preserve the phases of all participating modes.

Conceptual schematic. No computed root trajectory is shown.

3.4 Separate the three questions in a GBZ calculation

Constraint

Does the winding or minimum condition hold?

Spectral membership

Does the solution support zeros, or lie in a zero-free plateau?

Bloch-wave geometry

Which phases and connected point or line families survive?

Keep the full complex Bloch factors: the same decay rate can support different spatial oscillations.

Fixed-energy subsets become a sampled GBZ only after an energy sweep.

4.1 An analytic anchor: separable Hatano–Nelson

$$h = \sum_{j=1}^2 J_j (e^{\gamma_j} \beta_j^{-1} + e^{-\gamma_j} \beta_j), \quad J_j > 0$$

Axis-aligned open boundaries

$$\beta_j = e^{\gamma_j + i k_j}$$

$$E = 2J_1 \cos k_1 + 2J_2 \cos k_2$$

A diagonal similarity transform removes the hopping asymmetry.

Physical checks

Recover $\mu_j = \gamma_j$ and the real spectral interval $[-2(J_1 + J_2), 2(J_1 + J_2)]$.

Check the reconstructed mode families and their residuals under tighter tolerances.

Analytic benchmark, real J and γ in the stated basis. No numerical data are plotted.

4.2 What numerical evidence should test the bridge?

Physical question	Required data	Comparison
Localization	Model and complex Bloch factors	Exact benchmark; residuals
Geometry dependence	Same energy grid, chosen strip bases	SGBZ versus amoeba support
Mode families	Roots at fixed E and radius	Events and connected branches
Numerical reliability	Repeated tolerance/grid runs	Stable support; explicit failures

Select the Hamiltonian, parameters, lattice basis, diagnostic energy/radius, and energy grid before producing these figures.

DATA_REQUIREMENTS.md; collect_slide_data.py and plot_slide_data.py produce real-data SVG/PDF figures.

4.3 From roots to non-Bloch band geometry

Immediate output

Spectral support and the complex momenta of the selected mode families.

Compare decay lengths, phases, and their dependence on strip direction.

Next physical application

Reconstruct connected bands across energies, then study non-Bloch band geometry.

Eigenvector observables additionally require the Hamiltonian and left/right eigenvectors.

Band reconstruction remains an experimental extension; the characteristic polynomial alone does not determine eigenvectors.

4.4 The central message

Physical GBZ prescriptions become numerically accessible through polynomial roots.

Amoeba

Ronkin integral $\rightarrow$ root moduli $\rightarrow$ root counts $\rightarrow$ minimum condition.

SGBZ

Strip boundary problem $\rightarrow$ middle-root balance $\rightarrow$ loop winding $\rightarrow$ radius selection.

The project shares the root representation and preserves the distinction between the two physical formulations.

4.5 References

H.-Y. Wang, F. Song, and Z. Wang, **Amoeba Formulation of Non-Bloch Band Theory in Arbitrary Dimensions**, Phys. Rev. X **14**, 021011 (2024).

<https://doi.org/10.1103/PhysRevX.14.021011>

C. Wang et al., **Universal theory for geometry-dependent non-Hermitian bands**, arXiv:2506.22743v2 (2025).

<https://arxiv.org/abs/2506.22743v2>

Derivation assumptions, source mapping, and numerical data requirements accompany the editable deck.